

Glass-Core Advanced Packaging: updated case study anchored from 2025Q1

Glass substrates are best understood as a **manufacturing-bottleneck theme**: value shifts into the process stack that enables glass-core packaging once AI / HPC packages outgrow the limits of organic substrates.

Executive view

ARKA identified **glass substrates** not as a narrow materials trade, but as a migration of economics into the enablers of **through-glass-via formation, glass-compatible metallization, panel handling, inspection, metrology, and yield control**. The factor therefore tilts toward semicap, process control, and packaging-scale names that sit on the critical path if glass-core ramps selectively and slowly.

Investment thesis

Public validation already spans Intel’s glass-substrate roadmap, CHIPS/NIST support for Absolics’ glass-core ecosystem, Onto’s dedicated glass suite, Veeco’s AI-driven wet-process demand, and panel metrology / yield-control tools from FormFactor and PDF Solutions. The basket has tightened into **16 names** by 2026Q1, led by **ASX, IPGP, ONTO, FORM, VECO, and PDFS**.

FULL VALIDATION WINDOW

1.34 Sharpe

+101.8% total return · 45.2% ann vol
1 Jan 2025 → 20 Apr 2026

2026Q1 INFLECTION

+41.3 pts active

+36.7% factor vs **-4.6%** SPY
Strongest validation quarter in the rebased window

BENCHMARK READ-THROUGH

+81.0% active return

+1.62 information ratio · FF5 alpha **+39.7%**
Positive alpha; not fully style-explained

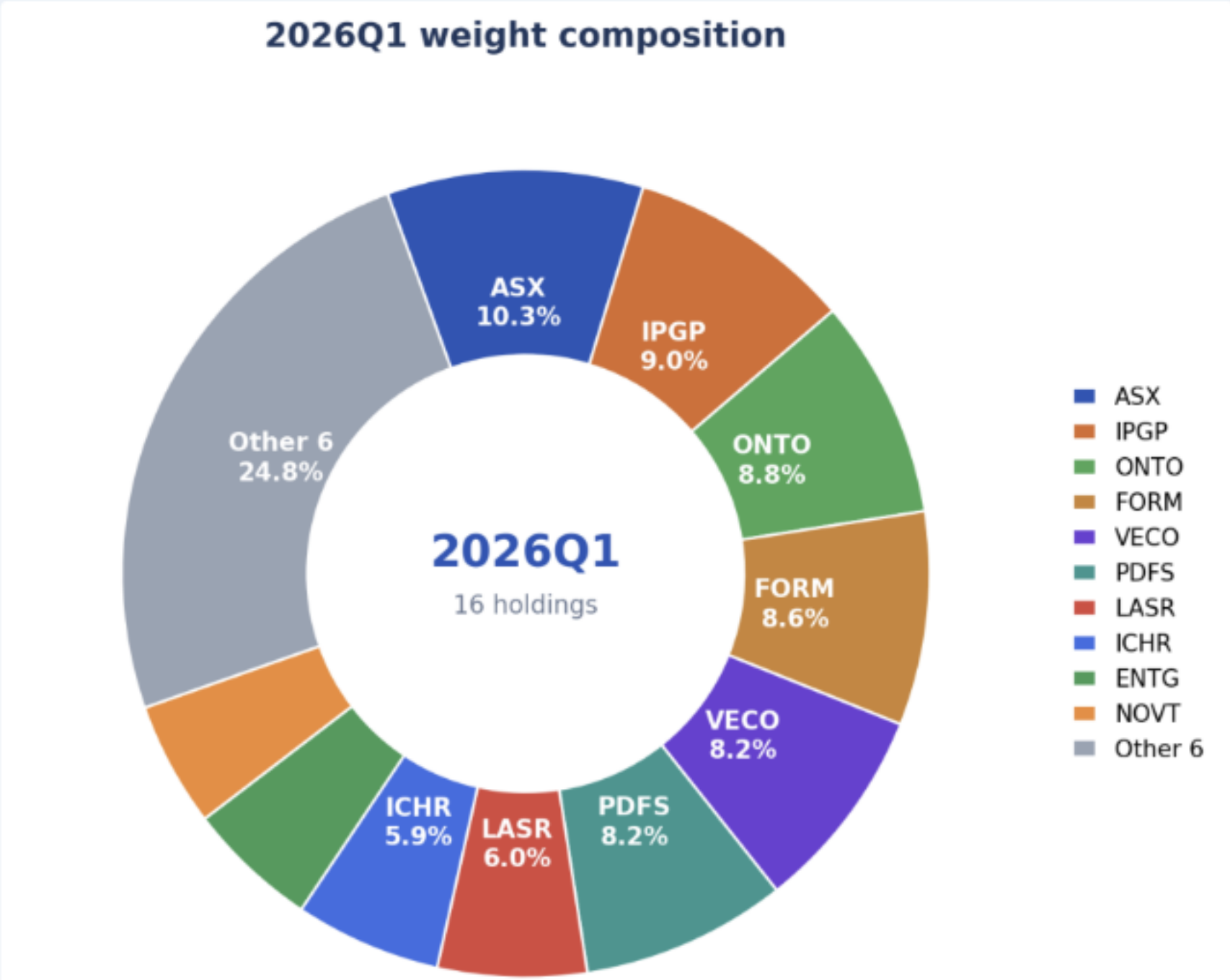
Indexed return since 1 Jan 2025 (rebased to 100)



2026Q1 holdings — full basket (16 names)

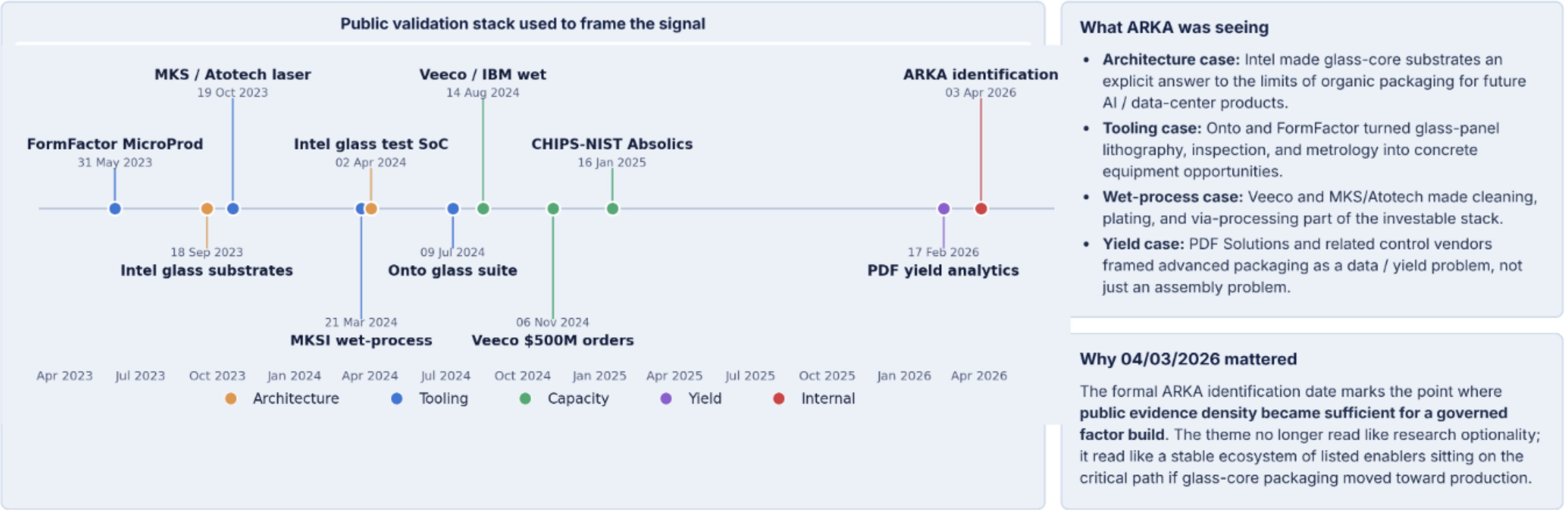
#	Ticker	Weight	Role in the basket
1	ASX	10.25%	Packaging-scale deployment / assembly
2	IPGP	9.04%	Laser source / via-formation read-through
3	ONTO	8.79%	Glass-panel lithography / inspection
4	FORM	8.56%	Panel metrology & inspection
5	VECO	8.25%	Wet processing / metallization
6	PDFS	8.23%	Yield analytics / manufacturing intelligence
7	LASR	5.97%	Laser micro-machining adjacency
8	ICHR	5.86%	Fluid delivery / chemistry infrastructure
9	ENTG	5.24%	Materials handling / plating / contamination control
10	NOVT	4.99%	Motion / sensing / automation
11	COHU	4.77%	Test / inspection / back-end quality
12	AMKR	4.52%	Advanced packaging / OSAT scale
13	ESI	4.41%	Specialty chemistry / metallization
14	AEIS	4.31%	Precision power / process control
15	MKSI	3.74%	Via drilling + chemistry / plating stack
16	KLIC	3.07%	Bonding / advanced assembly

2026Q1 weight composition



How the theme was identified — and why 04/03/2026 was the right identification date

The signal was **not a single substrate headline**. ARKA saw a convergence of architecture evidence, ecosystem funding, dedicated glass-panel tooling, wet-process demand, and rising value in inspection and yield control.

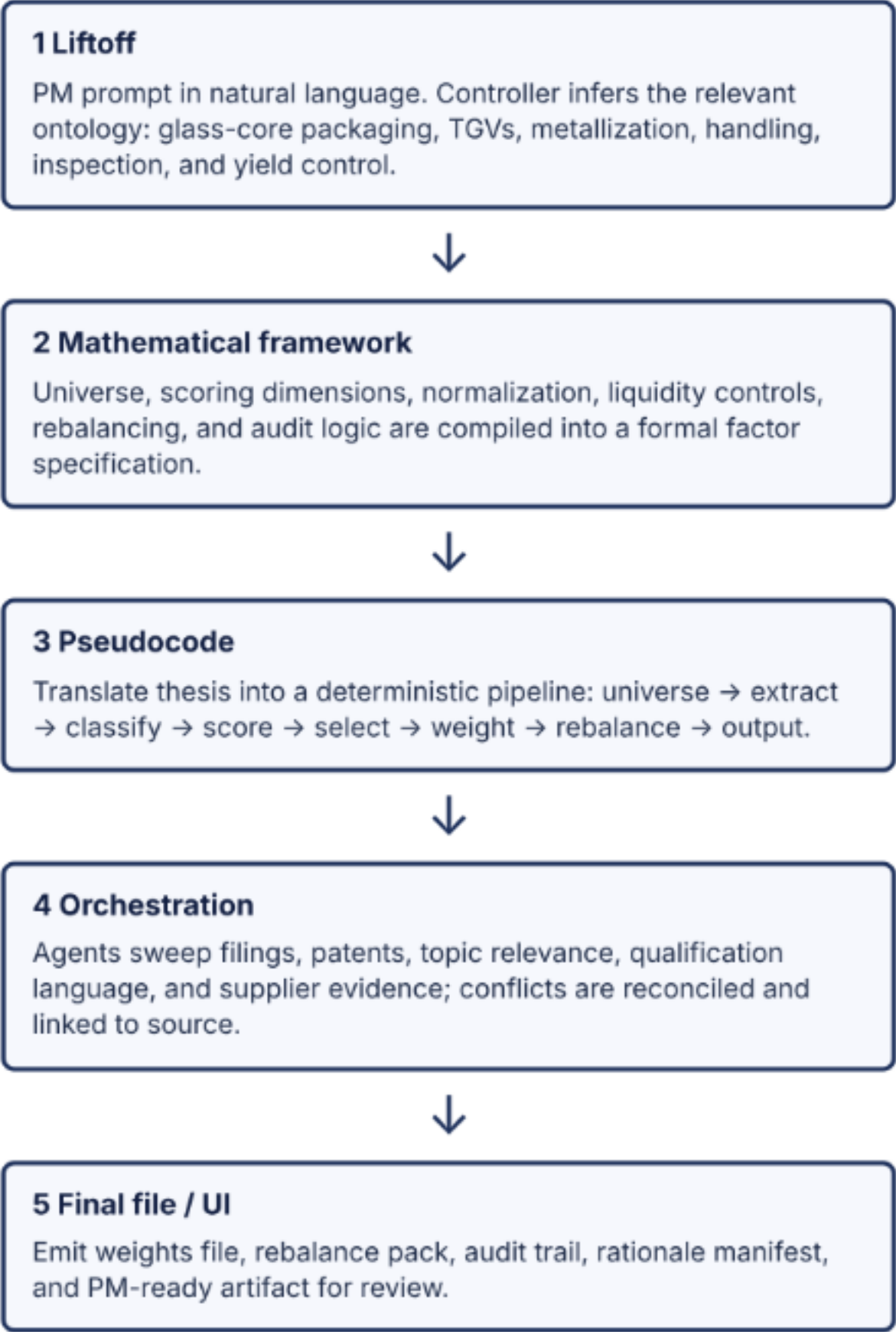


Classified trigger stack used to frame the signal

Date	Trigger	Type	Significance	Validation	Why it mattered	Ref.
31 May 2023	FormFactor launches MicroProf PT for panel metrology	Tooling / metrology	High	Official press release	Established panel-level metrology / inspection for large panels including glass.	P1
18 Sep 2023	Intel unveils glass substrates for AI/data-center packaging	Architecture validation	Very high	Official product architecture	Made the case that future AI/HPC packages outgrow organic materials.	P2
19 Oct 2023	MKS links ESI laser drilling with Atotech chemistry / plating	Process-stack validation	High	Official manufacturing solutions release	Showed that via formation and metallization can be sold as an integrated process stack.	P3
21 Feb 2024	Intel Foundry shows a glass-substrate test SoC	Foundry validation	High	Official foundry roadmap update	Moved the theme from concept to foundry-visible prototype evidence.	P4
9 Jul 2024	Onto launches a dedicated glass substrate suite	Inspection / lithography	Very high	Official product launch	Made glass-panel lithography, TGV metrology, and crack/chip inspection explicit equipment opportunities.	P5
14 Aug 2024	Veeco + IBM announce wet-processing collaboration	Qualification / ecosystem	High	Official joint-development agreement	Validated wet cleaning / strip / pre-bond steps as part of advanced-packaging adoption.	P6
6 Nov 2024	Veeco discloses >\$50m of AI-driven wet-processing orders	Commercial demand	Very high	Official order announcement	Converted advanced-packaging wet-process demand from roadmap into order flow.	P7
16 Jan 2025	CHIPS/NIST backs Absolics glass-core packaging ecosystem	Capacity / policy	Very high	Official U.S. government award	Created a domestic scale-up signal for glass-core packaging rather than pure lab research.	P8
13 Jan 2026	PDF Solutions frames advanced-package complexity in the AI era	Yield / data complexity	High	Official manufacturing commentary	Highlighted why yield-control and analytics become more valuable as advanced packages scale.	P9
17 Feb 2026	PDF Solutions pushes AI-driven collaboration and smarter decisions	Yield / analytics	High	Official analytics / platform commentary	Reinforced that manufacturing intelligence is a bottleneck-clearing layer, not a back-office function.	P10
3 Apr 2026	ARKA formal identification date	Internal synthesis	Very high	Controller trigger	Evidence density became sufficient for a governed factor build.	I1

Controller workflow: from prompt to portfolio-ready output

The attached framework is built for themes where direct revenue disclosures are sparse and the investable signal sits in **process-step language**. The controller converts a broad packaging thesis into explicit dimensions, a weighted score, capped quarterly weights, and an audit-ready handoff.



Universe selection can start from more than one anchor

For this factor the starting point is a U.S.-listed semicap / materials universe, but the controller is not forced into a single ontology. The PM can begin from a broad benchmark, a narrower semicap sleeve, or a targeted supplier pool and let the controller reconcile the union.

Core scoring equation

$$\text{CompositeScore}_i = 100 \times \text{logistic}(0.40 \cdot d_1 + 0.30 \cdot d_2 + 0.20 \cdot d_3 + 0.10 \cdot d_4)$$

After winsorization and sector-standardization; legacy exposure acts as a negative control dimension within the scoring stack.

Compiled exposure dimensions

Dimension	What it captures		Weight
D1	Glass-process revenue / backlog proxy	Estimated exposure to TGV formation, glass-compatible plating/metallization, panel handling, or glass-specific inspection tools	40%
D2	Design-in / qualification	Named customer activity, stage progression, design-ins, pilot / production language	30%
D3	Capacity / backlog proxy	Backlog mentions, capex commitments, customer orders, scale-up signals	20%
D4	Patent intensity	Recent relevant patents tied to TGVs, glass processing, plating, handling, or inspection	10%
Penalty	Legacy exposure drag	Negative weight on generic organic-substrate / legacy packaging language when glass relevance is weak	embedded

Controller pseudocode

Universe

Filter

Evidence

Classify

Score

Select

Weight

Maintain

Output

+ choose supplier universe relevant to advanced packaging / materials

+ tradability, listing, peer-group eligibility

+ filings + transcripts + patents + official product / order releases

+ glass revenue proxy, qualification, capacity/backlog, patents, legacy drag

+ winsorize → z-score within sector → weighted composite → logistic 0–100

+ keep names with positive thematic score and sufficient liquidity

+ proportional weights from composite score, then apply 8% name cap + 40% sector cap

+ quarterly rebalance; audit trail and rationale manifest at each cut

+ weights file + rebalance pack + PM review artifact

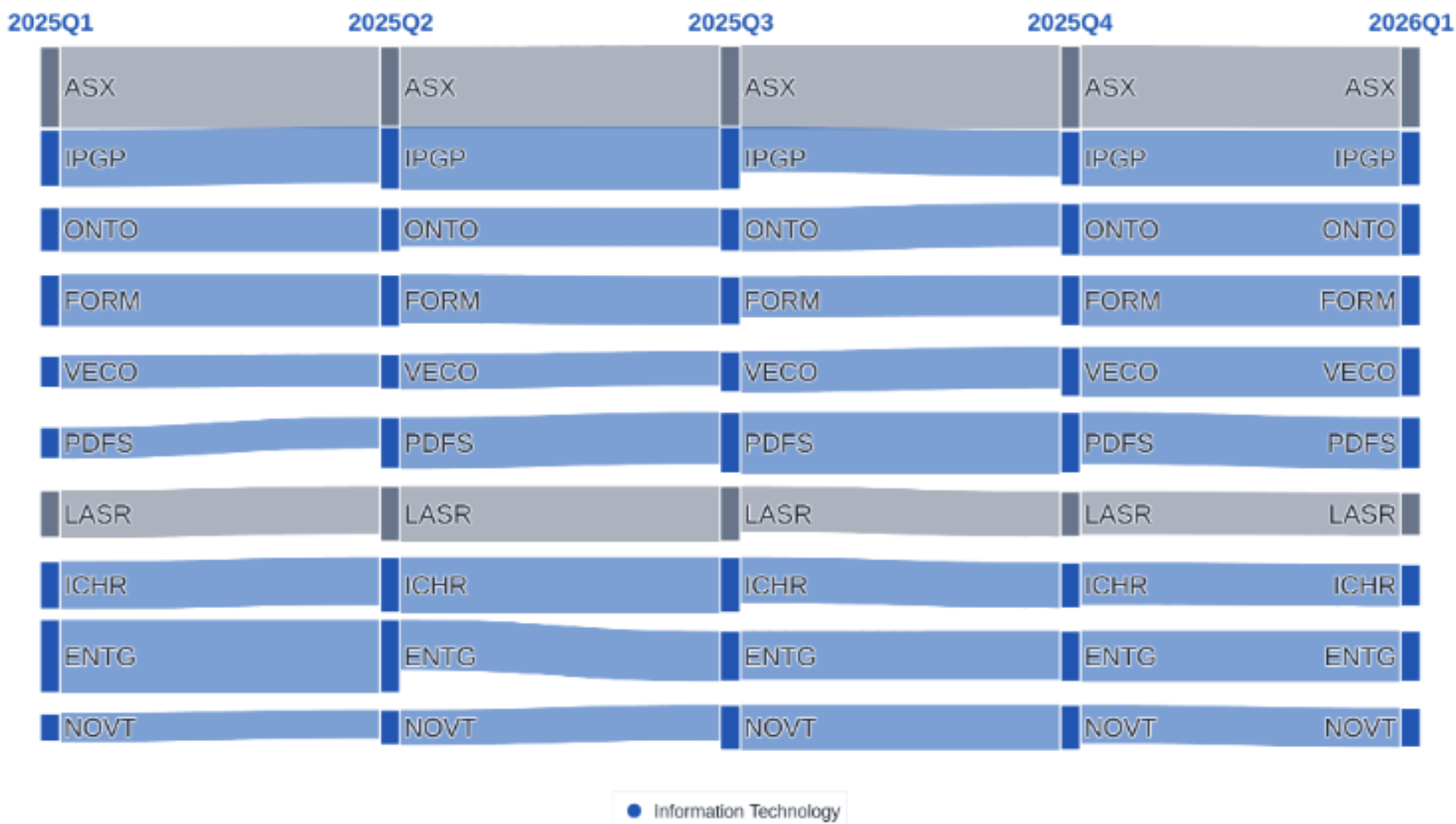
Freeform at inception → formal at scoring → deterministic at portfolio → auditable at output

Basket construction: explainable weight changes, not theme drift

The rebased window shows the basket tightening into the process stack that matters most if glass-core packaging is real, selective, and operationally difficult. The key point is **not concentration by itself** — it is that weight change follows validation, process priority, and portfolio discipline.

Top-10 weight migration of 2026Q1 constituents

Top-10 weight migration (2025Q1 → 2026Q1)



Concentration path since 2026Q1



What the weight changes are saying

- The chart is drawn only for names that survive into the **2026Q1 basket**, so it reads as a tightening of conviction rather than a full holdings census.
- Largest current weights are **ASX, IPGP, ONTO, FORM, VECO, and PDFS** — assembly scale, via-formation lasers, glass-panel inspection/lithography, wet process, and yield analytics.
- Into 2026Q1, the biggest gains versus 2025Q1 are in **IPGP, VECO, PDFS, FORM, LASR, ICHR, and ONTO**; the larger trims are **ENTG, AMKR, KLIC, AEIS, ESI, and MKSI**.
- The message is **not theme drift**; it is a reallocation toward the hardest-to-replicate bottlenecks under explicit portfolio caps.

Representative supply-chain map and why names matter

Stack layer	Representative names	Why weight moved / why it stayed
Laser / via formation	IPGP, LASR, MKSI	Laser-source, microvia drilling, and laser-enabled substrate processing tied to through-glass-via formation.
Wet process / metallization / chemistry	VECO, ESI, ENTG, ICHR	Wet cleaning, strip, chemistry delivery, contamination control, and plating / metallization infrastructure.
Inspection / metrology / yield control	ONTO, FORM, PDFS, COHU	Glass-panel lithography, TGV metrology, defect inspection, yield analytics, and back-end quality control.
Packaging-scale deployment / assembly	ASX, AMKR, KLIC	OSAT scale, advanced assembly, substrate handling, and bonding once the process stack becomes manufacturable.
Process-control adjacency	AEIS, NOV	Precision power, sensing, motion, and automation that support complex packaging flows.

Interpretation

The factor does **not** buy advanced packaging indiscriminately. It reallocates toward the steps that should capture more value if glass-core adoption is real but process-limited: **via formation, wet chemistry / metallization, panel inspection, metrology, and yield control**. That is why the basket's center of gravity sits in semicap, process control, and packaging-scale enablers rather than in broad semiconductor beta or legacy substrate incumbents.

Emergent Factor Intelligence for Institutional Portfolios

Performance diagnostics after rebasing to 2025Q1

The rebased window captures a difficult first quarter followed by a sequence of active-recovery periods. That makes the path more credible for PM review than a point-to-point thematic backtest.

Quarterly active-return profile

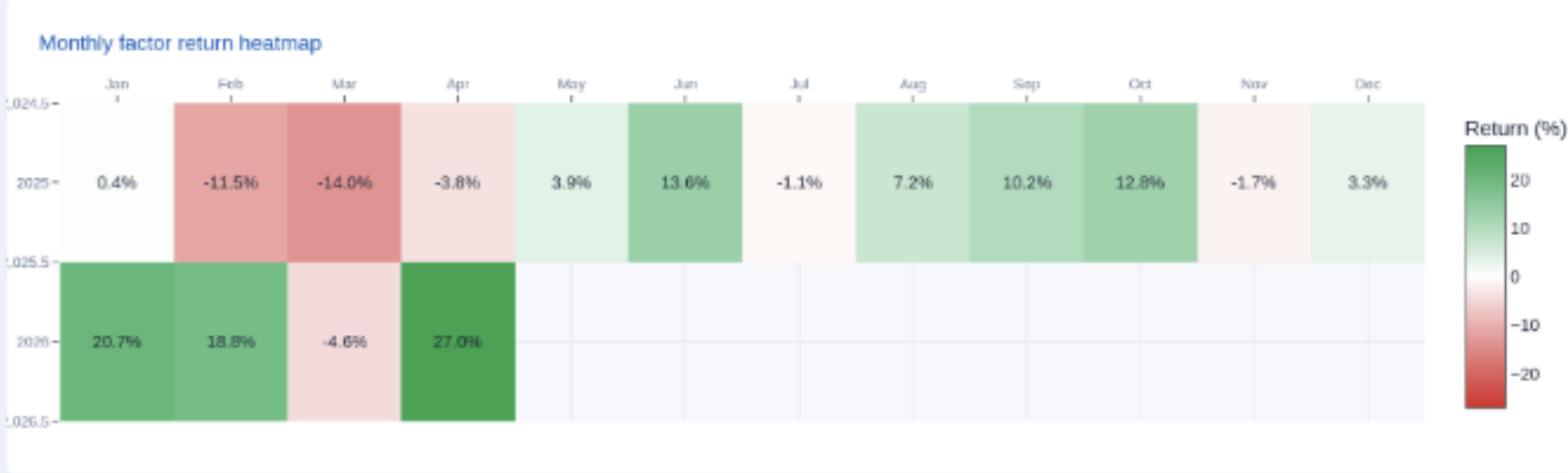
Quarterly returns — factor vs SPY (active) + diagnostics

Period	Factor	SPY	Shape	ModIO	Bonino	HE rate	Turnover	Active vs SPY
2025Q1	-23.8%	-4.3%	-3.87	-26.3%	-2.60	48.3%	—	-19.7%
2025Q2	+13.5%	+16.5%	1.85	-19.8%	1.53	52.8%	25.1%	+3.7%
2025Q3	+36.8%	+7.8%	1.88	-10.6%	4.48	53.1%	25.7%	+29.0%
2025Q4	+14.6%	+2.4%	1.45	-15.0%	2.36	58.9%	19.8%	+12.3%
2026Q1	+36.7%	-4.8%	2.86	-13.4%	8.18	62.3%	23.0%	+41.3%
2026Q2	+27.0%	+6.6%	12.98	-8.1%	14386.94	84.6%	—	+18.4%

Rolling benchmark sensitivity (63 trading days)



Monthly return heatmap (factor only)



POSITIVE MONTHS

62.5%

Share of months with positive factor return

OUTPERFORMANCE MONTHS

56.2%

Share of months beating SPY

MONTHLY UPSIDE CAPTURE

2.71x

Capture in up-SPY months

MONTHLY DOWNSIDE CAPTURE

0.83x

Capture in down-SPY months

BEST FULL MONTH

Apr 2026 +27.0%

Best complete month in-window

WORST FULL MONTH

Mar 2025 -14.0%

Worst complete month in-window

Updated read-through

The updated window still reads as a **pro-risk process-innovation factor**, but with two institutional strengths. First, the strategy absorbs a negative first quarter before later validation, which makes the path more believable than a monotonic backtest. Second, the strongest active returns arrive as the basket shifts toward the glass-process control points that matter most by late 2025 and early 2026, rather than simply loading broad semiconductor beta.

Distinctiveness, style read-through, and where returns came from

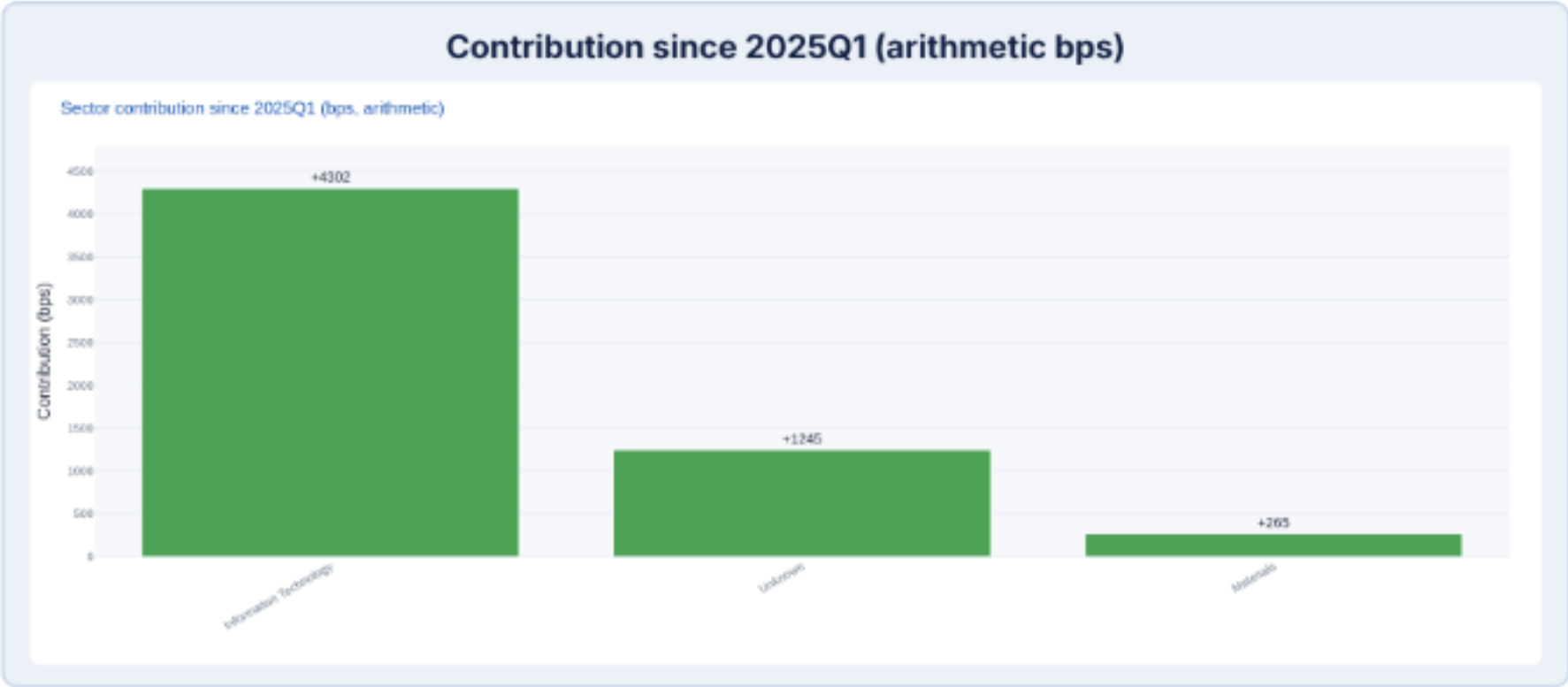
The updated window still reads as a pro-risk, stack-specific innovation factor. The point is not to claim market neutrality; it is to show that the basket is **not fully explained by conventional styles**.



FF5A archive reference (full attached history)

FF5 regression · $R^2 = 0.690$ · IR = +1.62

Factor	Beta	t-stat	p-value	Sig	Contribution (bps)
Mkt-RF	+1.64	+18.56	0.000	***	+1979.9
SMB	+0.88	+5.07	0.000	***	-254.6
HML	-0.48	-3.51	0.000	***	-544.1
RMW	-0.23	-1.49	0.137	n.s.	+152.3
CMA	+0.48	+2.27	0.023	**	+104.0
Alpha (ann.)	+39.75%	+1.63	0.104	n.s.	---



Updated interpretation

The factor carries meaningful **market beta (1.64)** and a positive **small-cap tilt (0.88)**, while leaning against value (-0.48) and quality / profitability (-0.23). **R^2 is 0.69**, so the correct institutional framing is “not fully style-explained,” not “orthogonal.” Annualized alpha is **+39.7%** with an information ratio of **1.62**.

Attribution and concentration

Since 2025Q1, the return path was still driven primarily by **Information Technology (+4302 bps)**, with meaningful additional contribution from the **Unknown / ADR bucket (+1245 bps)** and **Materials (+265 bps)**. Even so, concentration remained controlled: **top-5 weight 44.9%** and **HHI 0.07** in 2026Q1. The factor tightened into a more specific ecosystem without collapsing into a two-name trade.

Deployment scale and what this says about ARKA

The last test is implementation review. The attached archive tags this basket as **SMALL**, with the limiting ticker **PDFS** and total estimated trading cost of **5.7 bps** at **\$100M** reference AUM.

Illustrative deployment impact from the archive

Deployment impact curve — total trading cost vs AUM

AUM (USD millions)	Total cost (bps)	Market impact (bps)	Spread (bps)
100	5.7	0.5	5.2
1,000	13.0	1.0	12.0
2,000	16.5	1.5	15.0
3,000	19.5	2.0	17.5
4,000	22.5	2.5	20.0
5,000	25.0	3.0	22.0

Deployability snapshot

Deployability - estimated capacity 35.2M - tier SMALL

	AUM	Capacity tier	Impact (bps)	Spread (bps)	Total cost (bps)	Limiting name
	\$100M	SMALL	3.20	2.50	5.70	PDFS
	\$500M	SMALL	7.15	2.50	9.65	PDFS
	\$1.0B	SMALL	10.12	2.50	12.62	PDFS
	\$5.0B	SMALL	22.52	2.50	25.02	PDFS

Most capacity-sensitive current archive names

Most capacity-sensitive names (top 10) at \$100M AUM

	Ticker	Weight	ADV (USD mm)	% ADV at \$100M
	PDFS	8.23%	11.6	71.1%
	VECO	8.25%	34.8	23.7%
	COHU	4.77%	24.8	19.2%
	ICHR	5.86%	41.4	14.1%
	IPGP	9.04%	66.9	13.5%
	NOVT	4.99%	50.9	9.8%
	KLIC	3.07%	48.1	6.4%
	ASX	10.25%	174.9	5.9%
	FORM	8.56%	148.5	5.8%
	LASR	5.97%	108.0	5.5%

1 Not a static theme library

The prompt is freeform. ARKA first generates the relevant ontology, evidence stack, inclusion logic, and avoid flags for the thesis at hand.

2 Formal where it matters

Once the theme is compiled, the factor is constrained by explicit math: normalization, weights, caps, buffers, turnover guards, and liquidity review.

3 Self-adapting rather than fixed

Because the enablers of glass-core packaging are not a static ETF sleeve, the basket can reallocate toward the most validated bottlenecks as evidence converges.

4 Output is portfolio-ready

The endpoint is not a theme narrative. It is a weights file, rebalance pack, rationale manifest, and UI / report artifact PMs can review.

Freeform at inception. Disciplined in implementation. Repeatable across emerging manufacturing bottlenecks.